



Design, Implementation, and Performance Evaluation of Discrete and Three-Terminal Integrated Voltage Regulation Circuits

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Introduction

The stability of a power supply is a foundational requirement for the reliable operation of electronic systems. While most electronic devices require a constant direct current (DC) voltage, the raw output from rectified alternating current (AC) sources often contains significant voltage ripples and fluctuates as the load demand changes. Voltage regulation is the process of maintaining a steady output voltage regardless of variations in input voltage or the current drawn by the connected load (Boylestad & Nashelsky, 2012). This project explores the design and performance of two distinct methodologies; a discrete regulator and an integrated circuit (IC) regulator.

The discrete regulator design utilizes individual semiconductor components to achieve regulation, providing a practical look at the mechanics of power electronics. A high-power NPN transistor, the 2N3055, acts as the primary pass element, handling the bulk of the current and power dissipation (Onsemi, 2005). This configuration incorporates a TIP32C PNP transistor to function as a current limiter, which protects the circuit from overcurrent conditions by monitoring the voltage drop across a low-resistance 0.25Ω 5W cement resistor. The integration of large electrolytic capacitors, specifically a 4700 μF primary filter and 470 μF output smoothers, is critical for reducing the ripple voltage from the rectified input (Catalan, 2026).

In contrast, the IC-based regulator utilizes the LM317T, which integrates the error amplifier, voltage reference, and protection circuitry into a single TO-220 package (Texas Instruments, 2023). This approach simplifies the circuit layout while maintaining high precision through a feedback loop established by a fixed resistor network consisting of a 240Ω 0.5W resistor and a $1.5\text{k}\Omega$ 2W resistor. Despite the differences in complexity, both designs share a fundamental need for thermal management. Because linear regulators operate by dissipating excess voltage as heat, the use of specialized aluminum heatsinks, such as the EHS-08 for the 2N3055 and the EHS-81 for the TO-220 components, is mandatory to prevent thermal failure (Sedra & Smith, 2014).

The primary aim of this project is to bridge the gap between abstract power electronic concepts like diodes, transistors, and power supply concepts and practical hardware implementation. To achieve this, the following specific objectives and goals were established to guide the design, construction, and testing phases:

- Construct both a discrete regulator using individual transistors and an integrated circuit-based regulator using the LM317T.
- Ensure both designs provide a consistent 9V level, even when the input fluctuates or the load resistance changes.
- Verify the effectiveness of current-limiting mechanisms and thermal management using aluminum heatsinks and thermal paste to prevent component failure.
- Measure performance in the lab using variable resistance loads to determine how well the voltage holds steady under stress.

- Identify potential failure points and develop a practical understanding of how high-current conditions affect semiconductor junctions.

These goals directly align with the observed results, wherein the ease and exactness of the integrated circuit was contrasted against the implementation and its complexity of the discrete system, providing a complete educational perspective for voltage regulation.

Concept Theory

Voltage regulation is the process by which an electronic circuit maintains a constant output voltage (V_{out}) despite fluctuations in the input voltage (V_{in}) it changes in the current demanded by the load (I_L). In a linear regulator, the control element, which is typically a transistor, functions as a variable resistor in series with the load. By continuously adjusting its internal resistance through a feedback loop, the regulator compensates for variations to provide a stable output (Boylestad & Nashelsky, 2012). This process inherently involves *dropping* the difference between the input and output voltage ($V_{in} - V_{out}$), which is dissipated as heat.

A discrete regulator utilizes individual semiconductor components to establish a closed-loop control system. The 2N355 NPN power transistor acts as the main conduit for current. It operates within its active region, where its collector-emitter voltage (V_{CE}) is modulated to maintain the target output level. According to semiconductor theory, the output voltage is approximately $V_{out} = V_B - V_{BE}$, where V_B is controlled base voltage and V_{BE} is forward bias voltage drop of the silicon junction, typically 0.7V (Onsemi, 2005). As load current flows through the 0.25Ω cement resistor, it creates a proportional voltage drop. When this drop reaches the threshold voltage required to forward bias the TIP32C PNP transistor, the TIP32C begins to conduct. This divert base current away from the 2N3055, effectively throttling the output to prevent thermal failure or component damage during an overload or short circuit (Sedra & Smith, 2014). To maintain a precise output, the circuit requires a stable reference. Power diodes like the 1N5402, when forward biased, provide a predictable voltage drop. By comparing the output voltage to this reference via the transistors base, the circuit can automatically correct for deviations (Catalan, 2026).

The LM317T represents an evolution from discrete designs by integrating the reference, error amplifier, and pass transistor into a single package. The LM317T maintains a precise internal reference voltage (V_{ref}) of 1.25V between the Output (OUT) and Adjustment (ADJ) pins (Texas Instruments, 2023). The output voltage is determined by an external voltage divider. A 240Ω resistor (R_1) is placed across the reference pins to establish a constant program current. The second resistor (R_2), in this case a 1.5kΩ carbon film resistor, then sets the final output voltage according to the formula:

$$V_{out} = 1.25V(1 + R_2/R_1) + I_{adj}R_2$$

Since the adjustment pin current (I_{adj}) is negligible, the output remains highly stable and predictable (Texas Instruments, 2023).

Rectified AC power contains *ripple*, or periodic fluctuations in voltage. The inclusion of a large 4700 μ F electrolytic capacitor serves as a reservoir. During the peaks of the AC cycle, the capacitor charges, and during the troughs, it discharges to provide current to the load. This smoothing action provides a cleaner DC input for the regulator, which is further refined by smaller 470 μ F output capacitors that handle high-frequency noise and transient load changes (Boylestad & Nashelsky, 2012).

In linear regulation, power is wasted according to the equation $P = (V_{in} - V_{out}) \times I_{out}$. Thermal theory dictates that this power must be dissipated to keep the junction temperature (T_j) below the critical threshold. Aluminum heatsinks, such as the EHS-08 and EHS-81, increase the effective surface area for heat convection. Lowering the total thermal resistance from the junction to the ambient air is necessary to maintain the components within their Safe Operating Area (SOA) (Catalan, 2026).

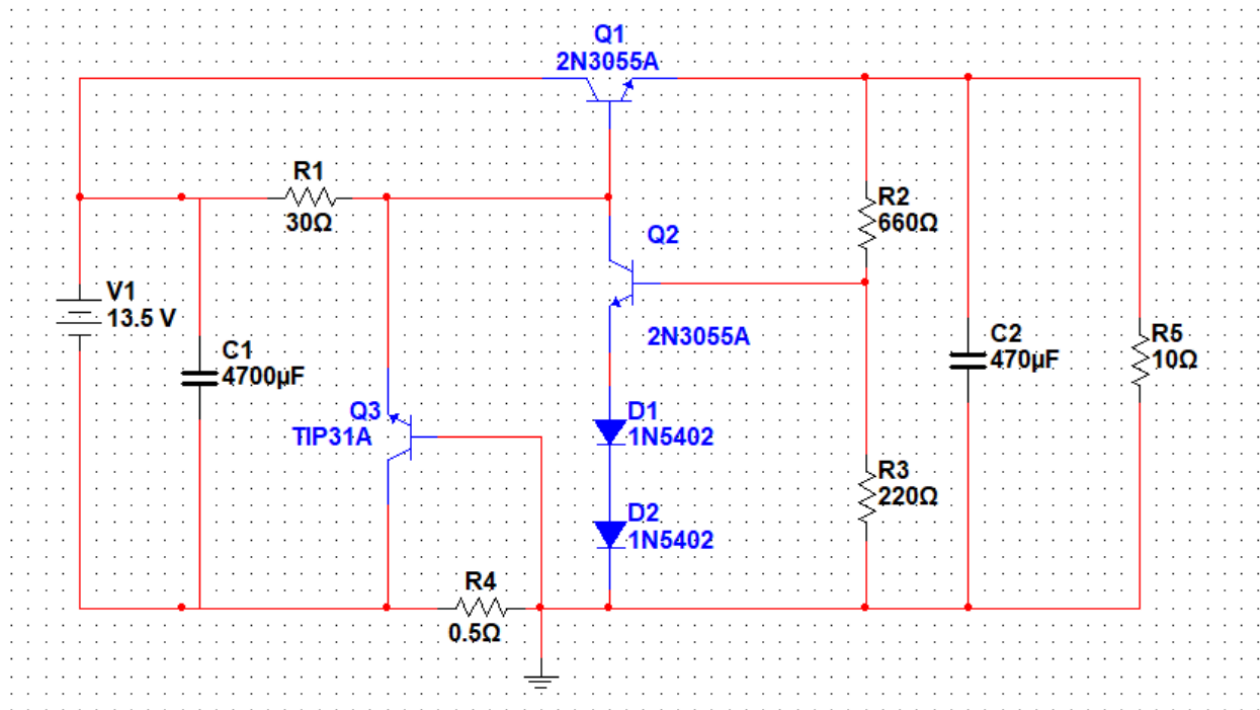
Design

3.1 Discrete Voltage Regulator

The design of this discrete voltage regulator focuses on achieving a stable 9V DC output at a maximum load of 1A. Our primary objective was to balance theoretical precision with the practical limitations of component availability and thermal constraints. The following details the calculations and the specific hardware decisions made to ensure the circuit survives real-world testing.

Figure 1.

Discrete Voltage Regulator Circuit Schematic



We utilized a series-pass regulator topology employing Darlington configuration Q1 and Q2 for high current gain and an NPN-based current-limiting mechanism Q3. The most critical design choice was the optimization of the input voltage V_{in} . While an initial calculation suggested a higher nominal input, thermal considerations led us to settle on 13.5V.

At $V_{in} = 13.5V$ and $V_{out} = 9V$, the power dissipation on the main pass transistor Q1 is:

$$P_D = (V_{in} - V_{out})I_{o(max)} = (13.5V - 9V)1A = 4.5W$$

This is a manageable load for the 2N3055 when mounted on our aluminum heatsink, staying well within the 60°C safety limit. We utilized the 2N3055 (NPN) for both the main pass transistor Q1 and the driver transistor Q2. While Q1 handles the bulk of the current, the low R1 value 30Ω

puts significant current stress on Q2, necessitating careful monitoring. A TIP32C (PNP) was selected for the current limiter transistor. The choice of a PNP transistor is essential here to implement high-side sensing. By placing the sense resistor R4 in the positive path, the TIP32C activates when the differential voltage across R4 exceeds the PNP's $V_{be(on)}$ threshold.

For the voltage divider, we transitioned from an initial 120Ω and 39Ω setup from our theoretical calculations to a more stable 660Ω and 220Ω configuration for R2 and R3 respectively. This ratio was calculated to set the output to 9V by comparing the feedback voltage to the $\sim 2.1V$ reference (two 1N5401 diodes + the V_{be} of the error amplifier). For biasing, we used a 30Ω (15W) resistance for R_I , achieved by series-connecting 3Ω (5W) cement resistors.

The practical implementation of the regulator shifted from temporary prototyping to a reinforced protoboard (perfboard) layout to ensure mechanical and electrical durability. To mitigate the risks of pad delamination and parasitic resistance associated with standard perfboards at 1A. The thermal strategy utilized a thick aluminum heatsink for the primary 2N3055 transistor. Because the TO-3 casing is electrically common with the collector V_{in} . Mounting the transistor directly to the metal without the thermal resistance of a mica insulator, which allowed for superior heat transfer, provided that the entire heatsink assembly remained electrically isolated from the system ground.

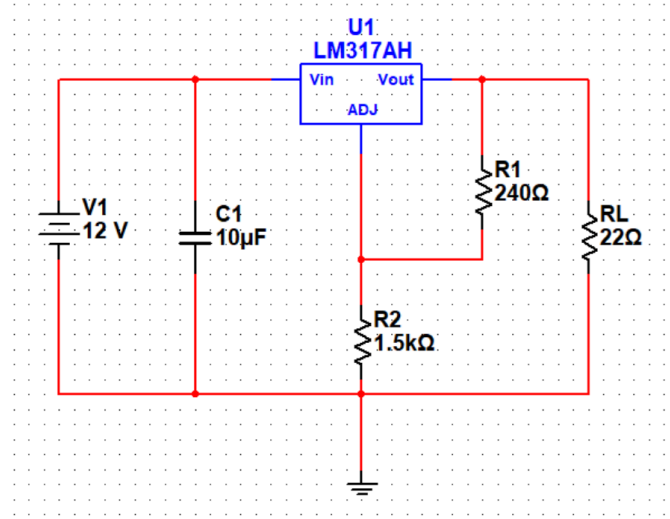
Verification of this design was centered around a custom-built 10Ω / 25W prebuilt test load, which provides a consistent stress environment to validate the regulator's load regulation and thermal stability within the assigned $60^\circ C$ operational limit. Actual design is that this test load is not directly soldered, but a variable load using a load box connected in place of R_L in actual regulation applications.

3.2 Three Terminal Voltage Regulator

The secondary phase of the project involved the design and implementation of a 3-terminal integrated regulator utilizing the LM317T variable voltage regulator IC. In contrast to the discrete circuit, this design leverages the internal architectural benefits of the integrated circuit, including built-in thermal overload protection and internal current limiting.

Figure 2.

Three Terminal Voltage Regulator Circuit Schematic



The design follows the standard application topology specified in the manufacturer's datasheet, which utilizes a reference voltage V_{ref} of 1.25V maintained between the Output and Adjustment terminals. To achieve the target 9V output, the resistive divider network was adjusted based on available high-tolerance components, specifically utilizing a 240Ω / 1/2W resistor for R_1 and a 1.5kΩ / 2W resistor for R_2 . This configuration was calculated using the standard transfer function:

$$V_{OUT} = V_{ref} \left(1 + \frac{R_2}{R_1} \right)$$

Resulting in a theoretical output of approximately 9.06V, which aligns with the observed stability requirements.

To enhance the stability and transient response of the regulator, a 10µF electrolytic capacitor C_1 was integrated at the input stage. This capacitor serves to bypass high-frequency noise and mitigate the effects of lead inductance from the DC source, ensuring a clean input for the LM317T's internal error amplifier and preventing potential oscillation. Furthermore, although the LM317T includes internal thermal shutdown features, a proactive thermal management strategy was prioritized to ensure reliability under continuous load conditions.

The TO-220 package was mounted to a dedicated heatsink using high-conductivity thermal paste to maximize the efficiency of heat transfer away from the junction. This decision ensures that the device operates within its safe thermal window, preventing the internal protection logic from prematurely throttling the output during rigorous load regulation testing. This implementation represents a streamlined, high-precision alternative to the discrete regulator, emphasizing integrated protection and ease of thermal management.

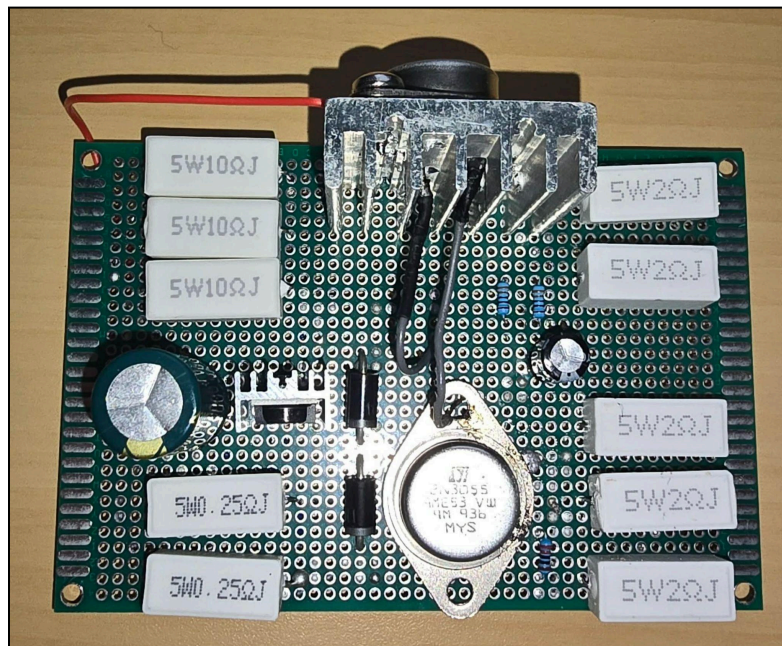
Methodology

4.1 Discrete Voltage Regulator

The design and implementation of the discrete voltage regulator circuit were carried out through a combination of analytical calculations, simulation-based validation, and practical hardware adaptation. The objective was to achieve a regulated output V_o of approximately 9V at a maximum load current $I_{o(max)}$ of 1A using a discrete transistor-based topology.

Figure 1.

Discrete Voltage Regulator Circuit Implementation



While initial theoretical models suggested a 13V nominal input, practical testing necessitated an increase to 13.5V to provide sufficient "headroom" for the Darlington pair to regulate while accounting for cumulative voltage drops across the junctions and the sense resistor.

Voltage Divider Optimization: A critical deviation occurred in the feedback network. The initial theoretical calculation used $R_2 = 120\Omega$ and $R_3 = 39\Omega$, yielding a ratio of approximately 0.245. However, this resulted in a high divider current:

$$I_{div} = \frac{V_o}{R_2 + R_3} = \frac{9}{159} \approx \mathbf{56.6mA}$$

To reduce thermal stress and quiescent power consumption, the divider was redesigned for higher impedance using $R_2 = 660\Omega$ (two 330Ω resistors in series) and $R_3 = 220\Omega$. This optimized network maintained the necessary comparison ratio while significantly lowering the current draw.

The biasing network was designed to ensure the series-pass element remains in the active region under full load. Theoretical requirements dictated a resistance of approximately 27Ω . Due to component availability and high power requirements, three 10Ω , 5W cement resistors were connected in series to obtain $R_1 = 30\Omega$. The maximum power dissipated across R_1 occurs during a short-circuit condition:

$$P_{D-R1(max)} \geq \frac{(V_{in(max)} - V_{be2(min)})^2}{R_1} = \frac{(13.5 - 0.6)^2}{30} \approx \mathbf{5.5W}$$

The selected 15W combined rating exceeds this requirement, providing a robust safety margin.

A sense resistor $R_5 \approx 0.5\Omega$ was selected to set the short-circuit current limit. At a maximum regulated current of 0.9A, the dissipation is:

$$P_{R5} = I^2 R = (0.9)^2 \cdot 0.5 \approx \mathbf{0.405W}$$

A 5W cement resistor was utilized to prevent thermal drift during current-limiting events. A TIP32C PNP transistor was chosen for the current-limiting role to implement high-side sensing, triggering a voltage foldback when the drop across R_5 exceeds the $0.7V V_{BE}$ threshold.

Thermal stability is the primary constraint for a high-current discrete regulator. For the primary 2N3055 transistor, the power dissipation is calculated as:

$$P_{D1} = (V_{in} - V_o) \cdot I_o = (13.5V - 9V) \cdot 1A = \mathbf{4.5W}$$

The dissipation for the driver stage was estimated to ensure it operates within safe case temperatures:

$$P_{D2} = (V_o + V_{BE1(max)}) \cdot I_{C2(max)} \approx (9 + 1)(0.836) \approx \mathbf{8.36W}$$

These calculations confirmed that while the main pass transistor handles the load, the driver stage Q2 remains a significant thermal concern.

Due to the high current levels, the design is to be a full protoboard (perfboard) implementation. The 2N3055 transistors were mounted on an aluminum bar heatsink using thermal paste. To

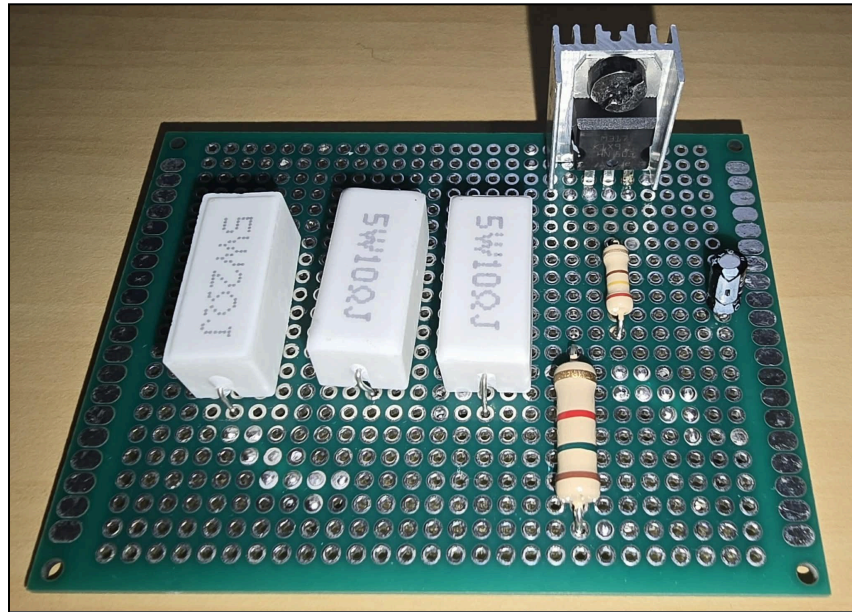
optimize heat transfer, a “mounted heatsink gorilla glued directly to the perfboard without touching any wires” strategy was used, continuity test was checked in all possible contact and it reported negative, mounting the TO-3 cases directly to the metal without mica insulators to reduce thermal resistance θ_{CS} , while ensuring the entire assembly was electrically isolated from the ground.

4.2 Three Terminal Voltage Regulator

The secondary phase of the project involved the implementation of a 3-terminal integrated regulator utilizing the LM317T IC. This design prioritizes efficiency and precision, leveraging the internal architectural advantages of integrated circuits such as built-in thermal overload protection and safe-operating-area compensation.

Figure 2.

Three Terminal Voltage Regulator Circuit Implementation



The design targets a stable 9V output V_o with a nominal load current of $0.4A I_{o(max)}$. To satisfy the dropout voltage requirements of the LM317T, the input voltage was established at $V_{in} \geq V_o + 3V$, resulting in a minimum required input of 12V. At these operating points, the output power P_{out} delivered to the load is calculated as:

$$P_{out} = V_o \times I_o = 9V \times 0.4A = 3.6W$$

The output voltage is programmed using an external resistive divider that references the internal $1.25V V_{ref}$ of the IC. Following standard design practices, R_1 was assumed to be 240Ω . The value for R_2 was determined using the transfer function:

$$V_o = V_{ref} \left(1 + \frac{R_2}{R_1} \right) \rightarrow 9V = 1.25V \left(1 + \frac{R_2}{240\Omega} \right)$$

Solving for R_2 yields 1488Ω , which was approximated to a standard $1.5k\Omega$ resistor. The power dissipation for these resistors P_{R1} and P_{R2} was verified to be $6.51mW$ and $40.04mW$ respectively, confirming that standard $1/4W$ resistors are sufficient for the feedback network. However, a $1/2W$ for $R1$ and $2W$ for $R2$ was chosen for extra safety.

For testing purposes, a 22Ω load resistor R_L was selected, resulting in an actual test current of approximately $0.414A$. The power dissipation of the load P_{RL} is $3.6W$, necessitating a $5W$ cement resistor to ensure safety during data collection. The load resistor is not permanently soldered into the circuit itself, and a variable load resistance is to be expected.

Thermal management for the IC is critical, as the power dissipated by the regulator itself P_{IC} is:

$$P_{IC} = (V_{in} - V_o)I_o = (12V - 9V)(0.4A) = \mathbf{1.2W}$$

Without a heatsink, the junction temperature T_j would reach approximately $95^\circ C$, based on an ambient temperature T_A of $35^\circ C$ and a θ_{JA} of $50^\circ C/W$. To maintain a safer operating temperature and prevent the internal thermal protection from throttling the output, the LM317T (TO-220 package) was mounted on a dedicated heatsink using high-conductivity thermal paste.

The circuit was implemented on a protoboard (perfboard) ensuring durability and accuracy. To stabilize the internal error amplifier and mitigate the effects of lead inductance from the $12V$ source, a $10\mu F$ electrolytic capacitor was integrated at the input stage. This combination of integrated protection logic and proactive thermal management results in a highly reliable and precise $9V$ regulation solution.

Testing Results

The testing of both voltage regulators performed using only no load and full load conditions was designed to evaluate the accuracy of the voltage output and the load regulation capabilities of both architectures under specific load conditions, without the use of laboratory equipment.

Table 1

Discrete Regulator and Three-Terminal Regulator Basic Testing

Parameter	Discrete Regulator	Three-Terminal Regulator
No-Load Voltage	3.12V	9.223V
Full-Load Voltage	~2.3V	9.224V
Maximum Current	0.41A	0.414A

The discrete regulator showed an output voltage that was significantly lower than the intended design target. Under no-load conditions, the circuit achieved a measured output of 3.12 V. When the full load was applied, the voltage decreased to approximately 2.3 V. The maximum current recorded during this testing phase was 0.41 A. These measurements indicate that the discrete regulator was unable to maintain the targeted nine-volt level during operation.

The three-terminal regulator exhibited a high degree of precision and stability throughout the testing process. The initial output voltage under no-load conditions was recorded at 9.223 V. Under full-load conditions, the output voltage remained remarkably consistent, with a measured value of 9.224 V. The system successfully delivered a maximum current of 0.414 A. These results confirm that the three-terminal regulator maintained a stable nine-volt level with negligible variation between load states.

The variable load test which was not documented by the group, however the group was able to test and demonstrate that it successfully regulates in the lab with the professor's checking. This is specifically only for the Three-Terminal Regulator, as the group has not attempted to present the discrete regulator.

Discrete Voltage Regulator Troubleshooting Steps

The implementation of the discrete voltage regulator was rigorous in terms of troubleshooting, moving through several critical stages of failure and adaptation, which eventually led to the identification of the problem the researchers cannot fix in time. Early testing showed significant output volatility. It was discovered that the reference diodes D_1 and D_2 were initially tied to a high-current ground path. This caused the reference voltage to float, leading to poor regulation. The circuit was re-wired using a Star Grounding strategy, moving the reference ground directly to the battery terminal to ensure a stable $\sim 2.1\text{V}$ comparison node.

The design utilized a 30Ω (15W) resistor array to provide the necessary base current for the Darlington pair. While mathematically sound for a 1A load, this low resistance provided a very high drive current to the driver transistor Q_2 . During a no-load calibration phase, Q_2 suffered a thermal failure. Without a load to draw current away from the pass element, Q_2 dissipated excessive power, resulting in the physical melting of the transistor junction. Post-failure testing showed the Q_2 collector clamped at 3.3V. This voltage collapse prevented the base of the pass transistor Q_2 from reaching the necessary $\sim 9.7\text{V}$ threshold required to sustain a 9V output, effectively "locking" the system at a 3V output level.

Further hardware modification was halted following a safety incident where the soldering iron cable was accidentally damaged and almost caused a live short circuit in a researcher's home.

Three-Terminal Voltage Regulator Troubleshooting Steps

The LM317T implementation successfully met all operational specifications, achieving a measured output of 9.10V. The troubleshooting phase for this circuit focused on optimizing the transient response and thermal overhead. To prevent the IC's internal thermal protection from throttling the output at the measured 0.414A load, the TO-220 package was mounted to a dedicated heatsink with high-conductivity thermal paste. This proactive thermal strategy kept the junction temperature well below the calculated uncooled limit of 95°C .

A critical component of the testing methodology was the implementation of a Star Ground for the LM317T's R_2 feedback resistor. Because the LM317T maintains a constant 1.25V reference between the Output and ADJ terminals, any parasitic voltage drop in the ground return path would shift the potential of the ADJ pin and introduce regulation errors. Furthermore, the data collection for the Load Regulation characteristics was conducted exclusively at the V_{OUT} terminal of the regulator rather than across the load resistor R_L . This decision was technically justified by the need to bypass the parasitic voltage drops associated with the interconnecting wires. At a test current of 0.414A, even minimal lead resistance would result in a measured voltage sag at the load that does not reflect the actual performance of the regulator IC. Measuring at the source ensured that the satisfaction of voltage output requirements was evaluated solely based on the circuit's internal regulation efficiency.

Discussion

The functional failure of the discrete regulator, characterized by the 3.3V clamp at the Q_2 collector, is primarily attributed to localized thermal runaway. While the series-pass transistor Q_1 was equipped with an aluminum heatsink, the driver transistor Q_2 was operated without external cooling. Under no-load conditions, the low resistance of the 30Ω bias resistor R_1 forced an excessive quiescent current through the Q_2 base-collector junction. In a Darlington configuration, the driver stage must be able to dissipate a portion of the total bias energy; however, the lack of thermal dissipation for Q_2 led to an internal silicon meltdown. This resulted in a permanent shift in the junction's resistive property, creating a "leak" to ground that effectively clamped the drive voltage. This failure illustrates that even in high-current designs, the bottleneck for reliability often lies in the lower-power driver stages if they are not appropriately de-rated or cooled for worst-case (no-load) scenarios.

The transition from erratic output to stable regulation in both circuits was achieved through the implementation of a Star Grounding strategy. In high-current circuits (1A), standard PCB traces or thin wires possess non-negligible parasitic resistance. If the reference nodes such as the LM317T's R_2 feedback resistor or the discrete regulator's comparison diodes are grounded along the same path as the load return current, a parasitic voltage drop is introduced.

This drop effectively shifts the "ground" potential as seen by the regulator, leading to "ground loop" errors where the circuit compensates for a voltage that does not exist at the output terminal. By utilizing a Star Ground, all reference components were tied directly to the battery's negative terminal, ensuring that the feedback loop functioned against a zero-potential reference unaffected by the 0.414A or 1A return currents.

A significant discrepancy was observed when measuring voltage at the Load Resistor R_L versus the Regulator Output Terminal V_{OUT} . This phenomenon is explained by Line Loss caused by the interconnecting wires. At a measured test current of 0.414A for the LM317T, even a lead resistance of 0.1Ω would result in a 41mV drop between the source and the load.

Because the regulator IC can only maintain its setpoint (9.10V) at its own pins, measuring at R_L would yield an inaccurately low reading, falsely suggesting poor load regulation. Relying on the V_{OUT} terminal for data collection was therefore necessary to isolate the performance of the regulator from the resistive losses of the external wiring.

The success of the LM317T implementation serves as a benchmark for integrated protection logic. Unlike the discrete circuit, which lacked thermal protection for the driver stage, the LM317T features internal circuitry that monitors junction temperature and current density. This allows the IC to throttle its output safely if thermal limits are approached.

Furthermore, the addition of the 10uF input capacitor was essential for mitigating high-frequency oscillations caused by lead inductance from the 12V DC source. These factors combined to produce a stable, precision-tuned output of 9.10V, demonstrating that integrated solutions significantly reduce the risk of catastrophic failure compared to uncompensated discrete topologies.

Conclusion

In conclusion, this project successfully demonstrates the fundamental principles of voltage regulation, emphasizing the ability of a circuit to provide a stable and consistent output voltage regardless of varying load resistance. Through the implementation of two distinct circuits, the discrete transistor-based regulator and the integrated three-terminal regulator. The group gained practical knowledge into the trade-offs between component-level engineering and integrated circuit convenience.

The integrated three-terminal regulator emerged as the most successful implementation, maintaining a precise output of approximately nine volts throughout the load testing phase. This stability was achieved through the use of a dedicated heatsink and proactive thermal management, which allowed the internal protection logic of the integrated circuit to operate without interruption. Furthermore, the addition of input filtering ensured that the circuit remained stable and free from high-frequency oscillations during high-current delivery.

In contrast, the discrete voltage regulator did not meet its primary objective of maintaining a nine-volt output during the final demonstration. Technical analysis unveiled that a critical component within the driver stage suffered a thermal failure during the calibration process, resulting in a degraded output voltage of approximately three volts. Although this prevented the circuit from meeting its quantitative targets, the subsequent troubleshooting process provided invaluable qualitative lessons regarding thermal budgets, the necessity of cooling for all active stages, and the physical limits of semiconductor junctions.

Ultimately, the results of this study highlight the critical role of precision in both design and measurement. The group identified that future projects would benefit from a more sophisticated understanding of parasitic resistance and grounding strategies, such as the use of a central grounding point to prevent reference shifts. By becoming more proficient in the use of measurement tools and more efficient in managing thermal dissipation, the accuracy and reliability of these regulated systems can be significantly improved in future engineering applications.

References

Boylestad, R. L., & Nashelsky, L. (2012). Electronic devices and circuit theory (11th ed.). Pearson Education.

Catalan, J. (2026). Basic electronics lab project design example: Discrete linear regulator circuit [Laboratory manual]. De La Salle University.

Onsemi. (2005). 2N3055 (NPN) / MJ2955 (PNP) complementary silicon power transistors [Data sheet]. <https://www.onsemi.com/pub/Collateral/2N3055-D.PDF>

Texas Instruments. (2023). LM317 3-terminal adjustable regulator [Data sheet]. <https://www.ti.com/lit/ds/symlink/lm317.pdf>

Sedra, A. S., & Smith, K. C. (2014). Microelectronic circuits (7th ed.). Oxford University Press.

Appendices

Appendix A

Material List and Price

LM317 Power Supply Regulator			
Components	Quantity	Price	Status
To-220 Aluminum Heatsink	1	36.15	Received
5W 2ohms Cement Resistor	5	114	Received
10pcs LM317T 2.2-37V	1	109	Received
10pcs 2W 1.5kohms Carbon Film Resistor	1	95	Received
20pcs 0.5W 240ohms Carbon Film Resistor	1	20	Received
6x8 cm Protoboard	1	77	Received
Discrete Power Supply Regulator			
Components	Quantity	Price	Status
2pcs 470uF 25V Capacitor	1	25	Received
1pc 4700uF 25V Capacitor	2	33	Received
2pcs TIP32C PNP Transistor	1	29	Received
2N3055 NPN 117W Transistor	2	33	Received
Thermal Paste	1	260	Received
0.25ohms 5W Cement Resistor	4	25	Received
10ohms 5W Cement Resistor	6	24	Received
Shipping Fee (PowerMav Electronics)	1	50	Received
8x12 cm Protoboard	1	79	Received
1N5402 (E-Gizmo)	5	42.5	Received
EHS-81 TO220 Heatsink (E-Gizmo)	1	8.5	Received
EHS-08 Heatsink (E-Gizmo)	1	15	Received
TOTAL			
LM317 PSR		451.15	
Discrete PSR		624	
Total		1075.15	

Appendix B

Discrete Voltage Regulator Initial Theoretical Calculations

$$I_{O \text{ max}} = 1A$$

$$V_{BE} = 0.65V$$

$$R_5 = \frac{V_{BE(Q5)}}{I_{O \text{ (max)}}}$$

$$R_5 = \frac{0.65V}{1A} = 0.65 \Omega$$

$$V_{base(Q3)} = V_2 + V_{BE(Q3)} = 8.2V + 0.65V = 8.85V$$

$$\frac{R_3}{R_2 + R_3} = \frac{8.85V}{9V} = 0.98$$

$$R_2 = \left(\frac{10000}{0.98} \right) - 1000 = 170 \Omega$$

$$I_{base(Q2)} = \frac{I_O}{\beta} = \frac{1A}{30} = 33.30mA$$

$$V_{R1} = V_{in} - (V_O + V_{BE(Q2)}) = 12V - (9V + 0.65V) = 2.35V$$

$$R_1 = \frac{2.35V}{0.05A} = 47 \Omega$$

$$P_{Q2} = (V_{in} - V_O) \times I_O$$

$$P_{Q2} = (12V - 9V) \times 1A = 3W$$

$$P_{out} = V_O \times I_O$$

$$P_{out} = 9V \times 1A = 9W$$

$$I_{load} = 1A$$

$$I_{bias(R1)} = 50mA = 0.05A$$

$$I_{total} = I_{load} + I_{bias} = 1.05A$$

$$P_{in} = V_{in} \times I_{total}$$

$$P_{in} = 12V \times 1.05A = 12.60W$$

$$P_{loss} = P_{in} - P_{out}$$

$$P_{loss} = 12.6W - 9W = 3.6W$$

Appendix C

Three Terminal (LM317) Voltage Regulator Initial Theoretical Calculations

LM317 Calculation:

Input Voltage Requirement

$$\begin{aligned} V_{in} &\geq V_o + 3V \\ V_{in} &\geq 9V + 3V \\ V_{in} &\geq 12V \end{aligned} \quad \left. \begin{aligned} V_o &= 9V \\ I_o &= 0.4A \end{aligned} \right\} \text{Group 1}$$

LM317 PD

$$\begin{aligned} P_{ic} &= (V_{in} - V_o) I_o \\ P_{ic} &= (12 - 9)(0.4) \\ P_{ic} &= 1.2W \end{aligned}$$

LM317 Thermal Consideration

$$\begin{aligned} T_j &= T_A + \theta_{JA} (P_{ic}) ; \theta_{JA} = 50^\circ\text{C/W} \\ T_j &= 35 + (50)(1.2W) \quad (\text{for LM317}) \\ T_j &= 95^\circ\text{C} \end{aligned}$$

Output Power

$$\begin{aligned} P_{out} &= (V_o)(I_o) \\ P_{out} &= (9)(0.4) \\ P_{out} &= 3.6W \end{aligned}$$

Resistors :

$$R_1 = 240\Omega \quad (\text{CWE assumed})$$

$$V_o = V_{ref} \left[1 + \frac{R_2}{R_1} \right] ; V_{ref} = 1.25V \quad (\text{for LM317})$$

$$9 = 1.25 \left[1 + \frac{R_2}{240\Omega} \right]$$

$$R_2 = 1488\Omega \approx 1.5K\Omega$$

Resistors PD

$$P_{R1} = \frac{V_{REF}^2}{R_1} = \frac{1.25^2}{240} = 6.51\text{mW} \rightarrow \frac{1}{4}W \text{ Resistor good}$$

$$P_{R2} = \frac{V_{R2}^2}{R_2} = \frac{(V_o - V_{REF})^2}{R_2} = \frac{(9V - 1.25V)^2}{1500} = 40.04\text{mW} \rightarrow \frac{1}{4}W \text{ Resistor good}$$

Load Resistor :

$$R_L = \frac{V_o}{I_o} = \frac{9V}{0.4A} = 22.5\Omega \approx 22\Omega \quad \text{but } 0.4A$$

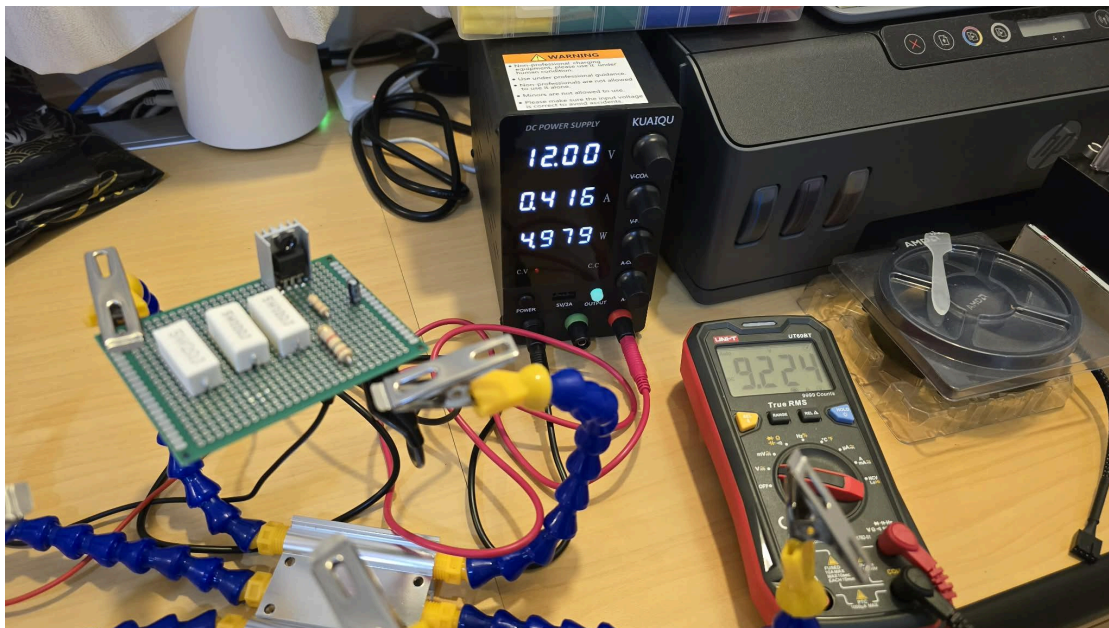
Load Resistor PD:

$$P_{RL} = V_o \times I_o = P_{out} = 3.6W$$

5W Resistor good

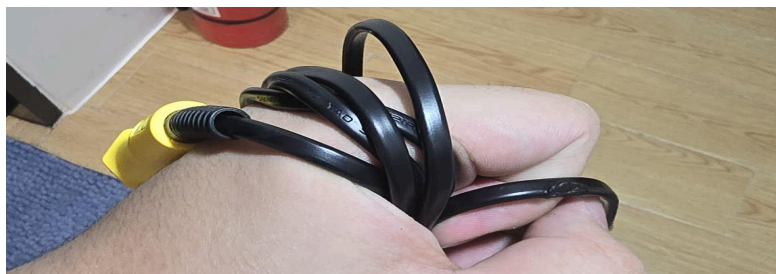
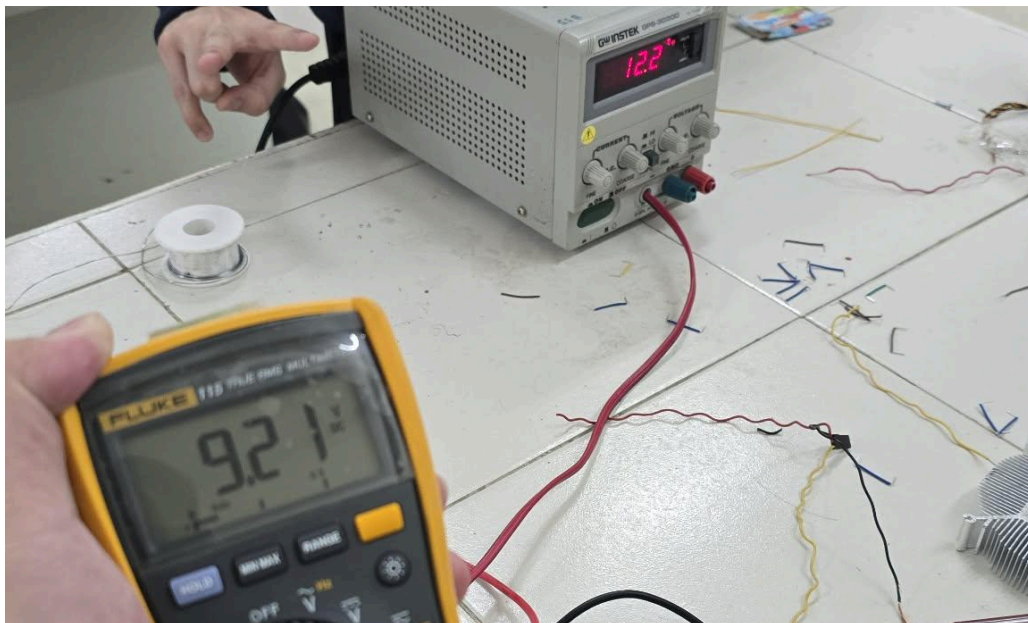
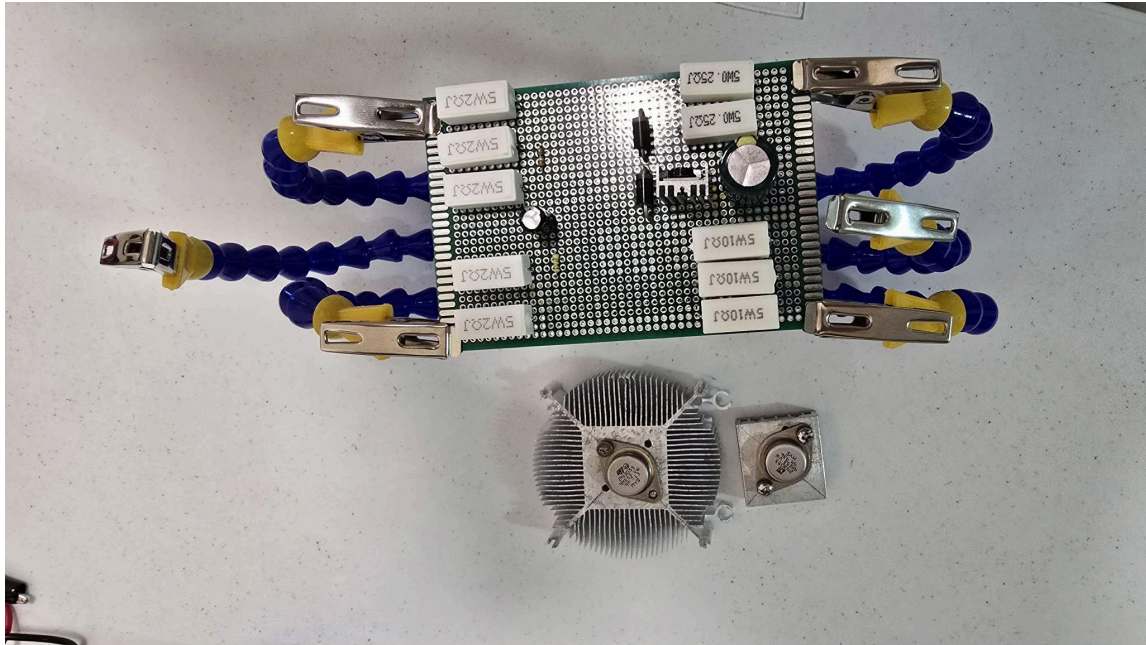
Appendix D

Three Terminal Voltage Regulator Test Documentation



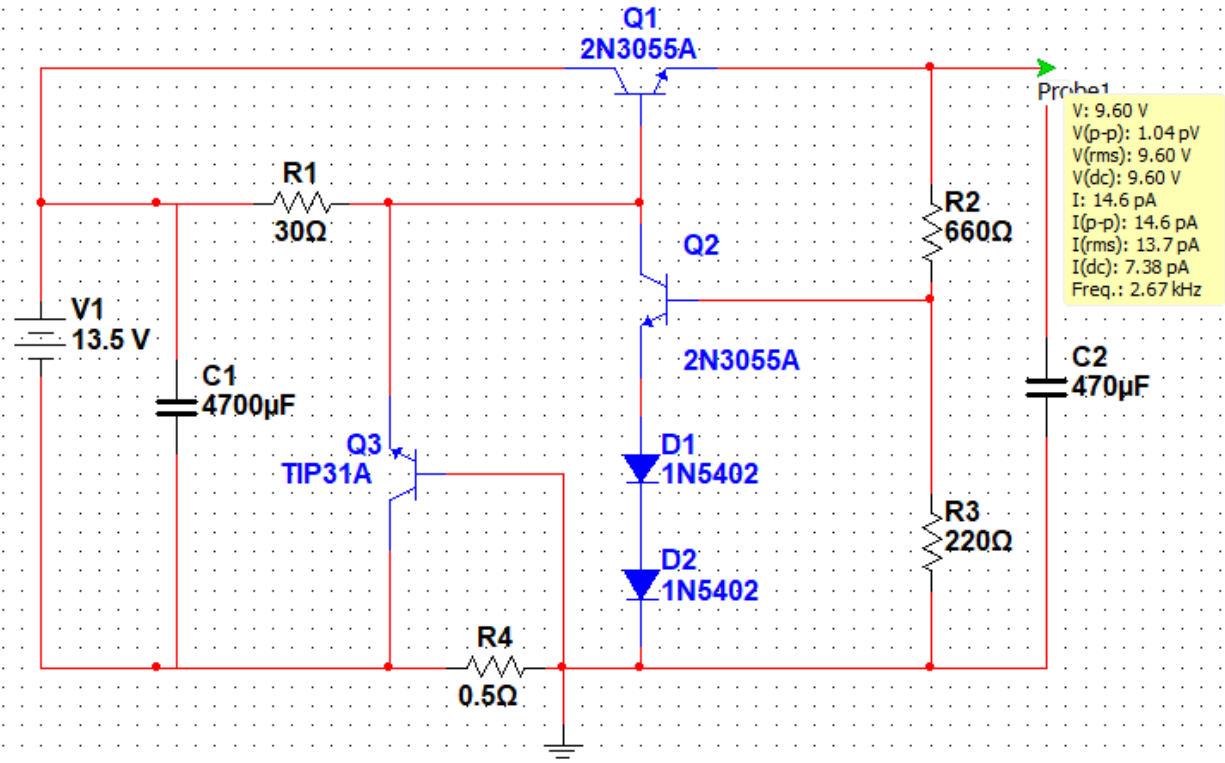
Appendix E

Voltage Regulators Process Documentation



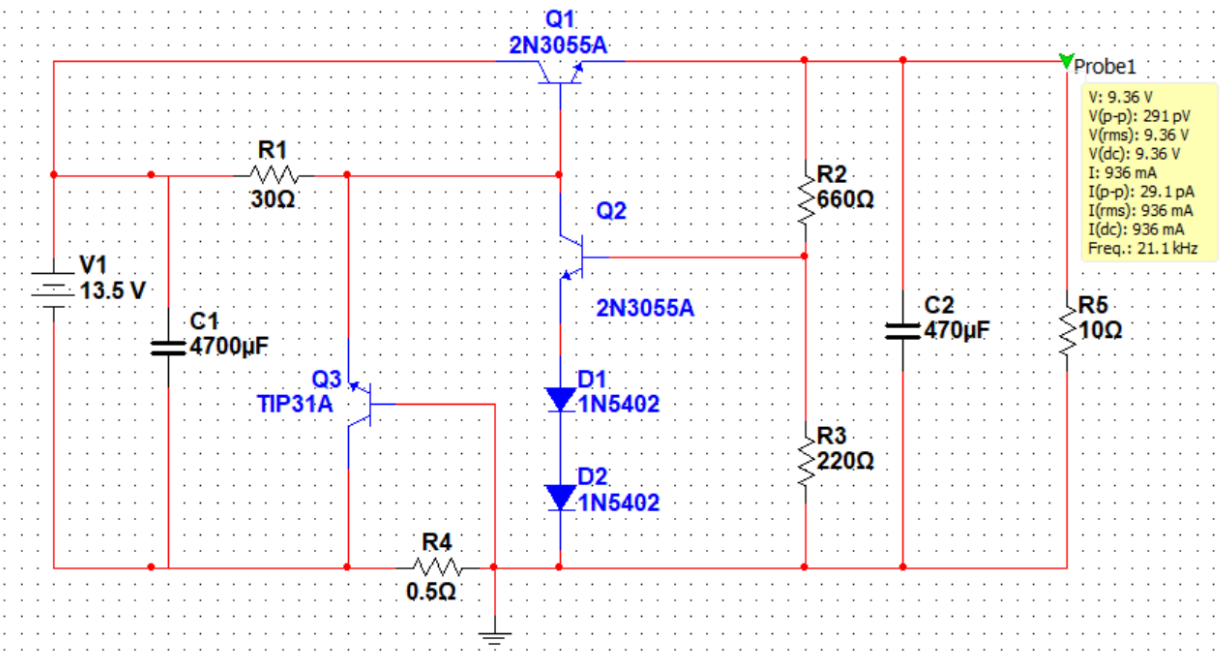
Appendix F

Discrete Voltage Regulator No Load Simulation



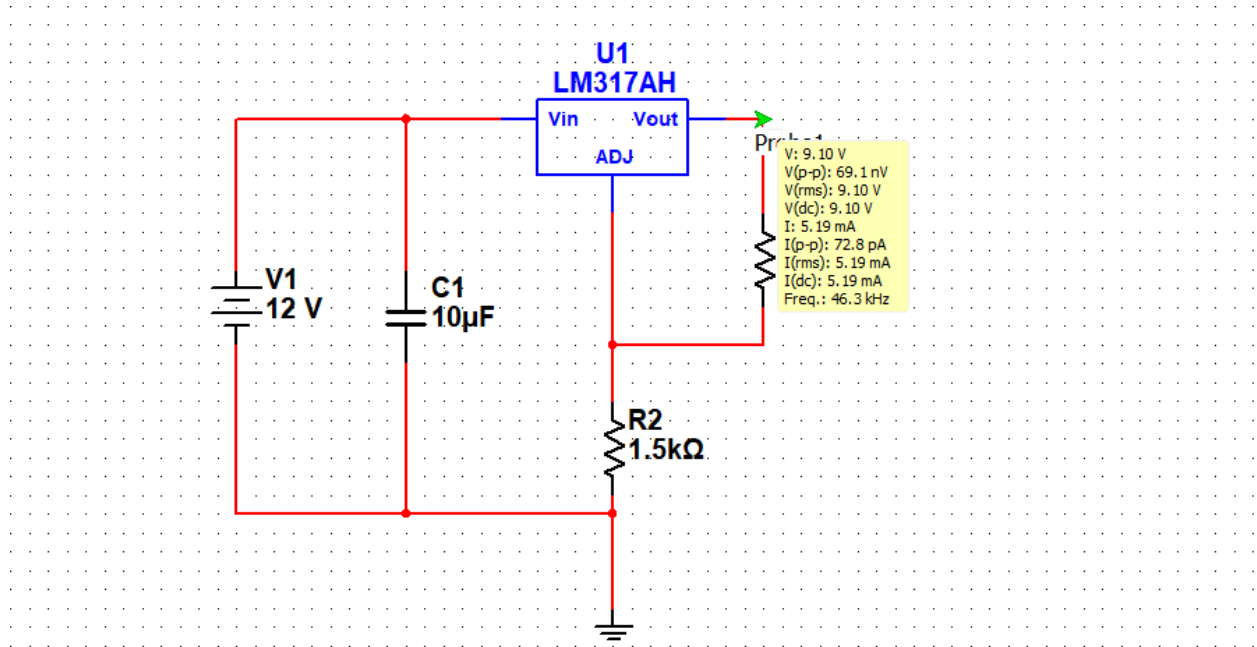
Appendix G

Discrete Voltage Regulator Full Load Simulation



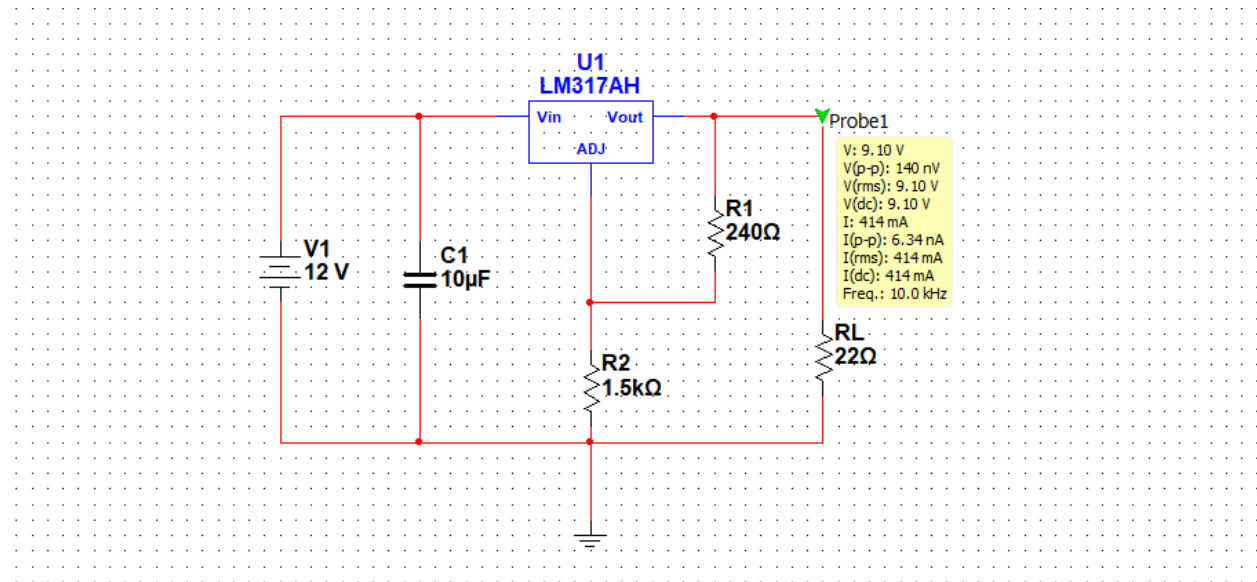
Appendix H

Three-Terminal Voltage Regulator No Load Simulation



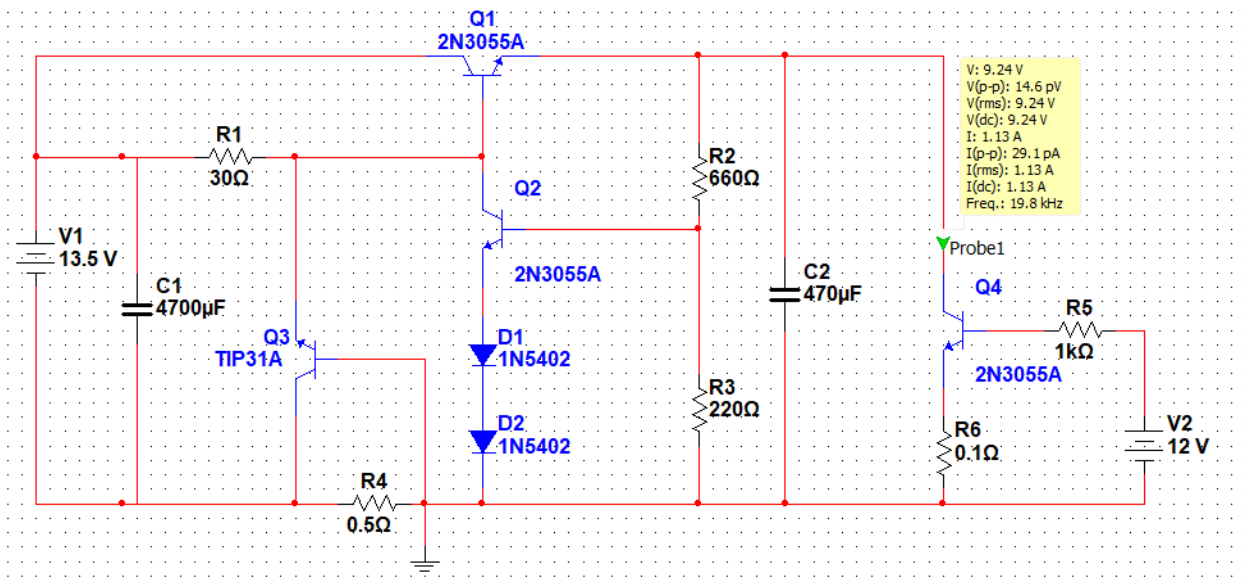
Appendix I

Three-Terminal Voltage Regulator Full Load Simulation



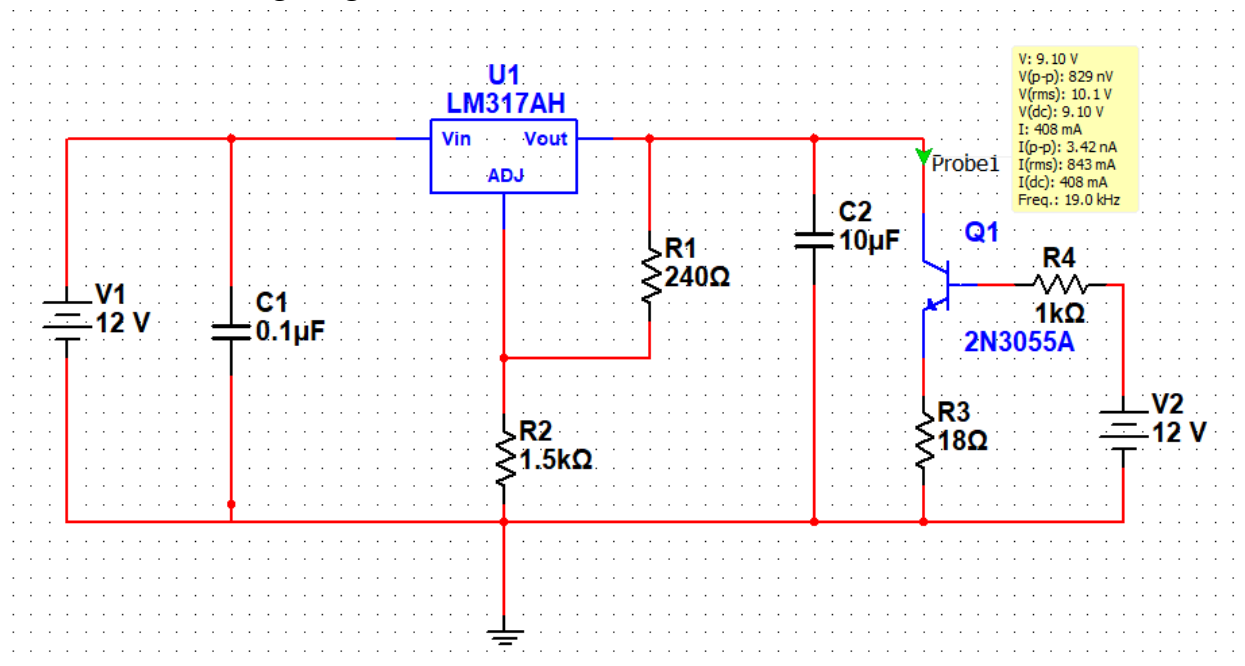
Appendix J

Discrete Voltage Regulator Variable Load Simulation



Appendix K

Three-Terminal Voltage Regulator Variable Load Simulation



Appendix L

Group Members After Documentation

